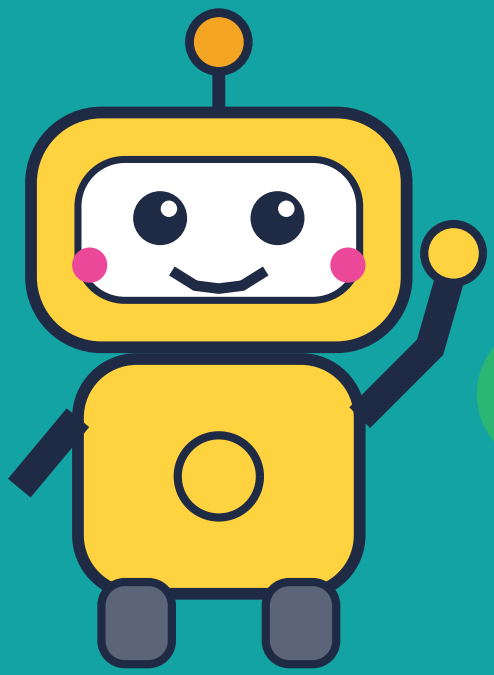
SPECIMEN BOOKLET

Think Like a Coder

Computational thinking for Classes 3–5

A sample for schools: three full chapters, a teacher script, the classroom kit and how assessment works



For CBSE's Computational Thinking and AI curriculum, 2026-27

Class 5 sample · Follows the NCERT maths chapters · No computers needed

Practice Pack Rs 449 per child (indicative) · Free pilot, January-March 2027

© 2026 Bullseye Knowledge System Limited. All rights reserved. Not for sale.
CodeSchool is a working name. Not an official CBSE or NCERT publication.

A complete CT&AI programme for Classes 3-5

From 2026-27, CBSE asks schools to teach Computational Thinking and AI in Classes 3-8: **50 hours a year in Classes 3-5**, built into Mathematics and The World Around Us periods. CBSE's free handbooks give puzzles for each maths chapter. CodeSchool adds everything a teacher needs to teach it well, with no computers needed.

What schools need	CBSE's free handbook	CodeSchool Practice Pack
Puzzles for each maths chapter	Yes	Yes, in levels from easy to challenge
Theory in simple words	Short introduction	A Learn page and Robo's tip in every chapter
Worked examples	No	One per chapter, step by step
Coding	No (follow steps only)	'Now you build it': coding on paper with code cards
Hands-on activities	One or two a year	A classroom kit used all year
Teacher lesson plans	Answer keys	A period-by-period teacher script book
Assessment and report card	No format given	Self-check, observation journal and term summary
Teacher training	CBSE workshops	School orientation, termly workshops and support

★ Why schools choose it

- ✓ Follows the chapters of the NCERT Class 5 maths textbook in the same order, so it fits the maths period.
- ✓ Built around the CT skills in CBSE's curriculum: steps, patterns, abstraction, decomposition, logic and data.
- ✓ Works in any classroom: paper, cards and a floor mat. A projector helps but is optional.
- ✓ Ready for a maths or EVS teacher, not only a computer specialist.

© 2026 Bullseye Knowledge System Limited

What each school receives

For	What they get	Classes
Each child	Practice Pack: level workbook with coding cut-outs and take-home challenges	3, 4, 5
Each class section	Classroom kit: code cards, floor grid mat, cipher wheels, sorting cards, box	3, 4, 5
Each teacher	Teacher script book, answer keys, observation journal, short how-to videos	3, 4, 5
The school	Orientation at school, a workshop every term, a support line and a term summary format	All

The series

Book	Levels	Coder skills
Class 3 Practice Pack	B1-B3	Steps and codes, views, loops, decomposition, patterns, events, data, logic, sorting, if-then, variables, grids
Class 4 Practice Pack	B3-B4	If-else choices, parity checks, sorting networks, nested and repeat-until loops, decision trees, data
Class 5 Practice Pack (this sample)	B5-B7	Variables, search strategies, debugging and a game designed on paper
Classes 6-8 Projects and AI Pack	C1-C8	Functions, data, the AI chapters and two 20-hour projects a year (separate sample)

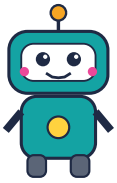
Price and ordering

Indicative price: Rs 449 per child per year (MRP) for the Classes 3-5 Practice Pack, through the school's book list. It can replace an existing computer book, so the cost to parents stays about the same. Kits, teacher books and training come with the packs. Prices are indicative until confirmed in writing.

Try it first

A **free pilot** in two or three sections from January to March 2027: materials, a teacher orientation, weekly support and a short report against success measures agreed in advance. No obligation to continue.

© 2026 Bullseye Knowledge System Limited



Hello! I am Robo. I think like a coder, and soon you will too! In this book we will solve puzzles, crack secret codes, guide robots and write programs, all on paper.

How this book works

Learn — Robo explains one big idea in simple words, with the new coder words to remember.

Example — A worked example shows the idea step by step.

Try it! — Five practice questions. Circle the answer or write it in the box.

Now you build it — Coding on paper: put code cards in order, fix bugs and write your own programs.

Challenge — A harder puzzle for when you are ready for more.

How did I do? — Colour a face to show how sure you feel.

My code cards

Coders use cards like these. Each colour means something different. You will find cut-out cards at the back of the book.

Step: do this | **Repeat: a loop** | **If: a decision** | **When: an event** | **move**

For teachers and parents

Each chapter goes with the same chapter of the NCERT Class 5 maths textbook, so it can be used in maths periods. CBSE suggests about 50 hours of computational thinking a year in Classes 3-5. The coding here needs no computer: children act out, write and check programs on paper, which builds the same ideas used later in block coding and AI. Answers are at the back. Encourage children to explain their thinking, not just the answer.

© 2026 Bullseye Knowledge System Limited

How a lesson runs

Every lesson fits one 40-minute period and follows the same five steps, based on the PRIMM approach used in computing classrooms (predict, run, investigate, modify, make). Children learn the routine once and then focus on the thinking.

1. **Predict and run** 8 min
   Children predict what a set of steps will do, then act it out.

2. **Investigate** 8 min
   Pairs answer 'why' questions about how it worked.

3. **Modify** 8 min
   Children change the steps to do something new.

4. **Put in order** 8 min
   Jumbled code cards go back in the right order.

5. **Make** 8 min
   Pairs write their own program and swap with another pair.

How it supports CBSE's assessment

CBSE asks schools to assess CT&AI through written puzzles, group activities and a **Teacher Observation Journal**. Each step gives the teacher something to observe: predictions show reasoning, modifications show understanding, and the 'make' step shows whether a child can build an algorithm of their own. The journal format and a term summary for the report card come with the teacher book.

Every child can take part

- ✓ Children who finish early move to the challenge level; children who need help use a hint card.
- ✓ Most activities are done in pairs, so children who find reading hard can still take part by talking and acting out.
- ✓ Teacher preparation is about 5-10 minutes per lesson using the script.

© 2026 Bullseye Knowledge System Limited

Inside the full Class 5 book

The complete book has 15 chapters, one for each chapter of the NCERT Class 5 maths textbook, plus answers, a glossary and cut-out code cards. This sample includes chapters 8, 9, 14 in full.

Ch	Chapter	Coder skill	Level	In this sample
1	We the Travellers - I	True, false or can't tell: logic with AND and OR	B4 review	
2	Fractions	Variables that keep score	B5	
3	Angles as Turns	Turtle turns: angles in degrees	B4	
4	We the Travellers - II	Inputs and variables: fuel left	B5	
5	Far and Near	Compare and decide	B5	
6	The Dairy Farm	Counters and running totals	B5	
7	Shapes and Patterns	Pattern generators: rules for any step	B4	
8	Weight and Capacity	Search strategies: halve it to find it fast	B6	Yes
9	Coconut Farm	Input-output rules (function machines)	B5	Yes
10	Symmetrical Designs	Design machines: turning symmetry	B4	
11	Grandmother's Quilt	Tiling and decomposition	B6	
12	Racing Seconds	Timers: variables that change over time	B5	
13	Animal Jumps	Jump rules: loops with conditions	B6	
14	Maps and Locations	Capstone 1: design a treasure-map game	B7	Yes
15	Data Through Pictures	Capstone 2: scores, data and improving the game	B7	

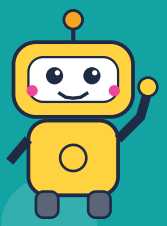
© 2026 Bullseye Knowledge System Limited

CHAPTER 8 · LEVEL B6

8 Weight and Capacity

Coder skill: Search strategies: halve it to find it fast

Goes with NCERT maths, Class 5, chapter 8

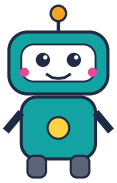


Learn

A **one-by-one search** (linear search) checks things in order until it finds the right one. It can take a long time.

A **halving search** (binary search) checks the middle and throws away the half that cannot hold the answer. For 1 to 32 it needs at most 5 yes/no questions. Computers use it to find things very fast.

linear search	halving search	strategy
check one by one	check the middle, drop half	a plan for solving



To guess a number from 1 to 32, I always pick the middle. Each guess throws away half the numbers!

Example

Guess my number (1 to 32)

1. Guess 16: 'higher'. Now only 17 to 32 are left.
2. Guess 24: 'lower'. Now only 17 to 23 are left (grey numbers are ruled out).
3. Keep guessing the middle of what is left. You will find it in a few more guesses.



★ *Coder fact!* Phone and dictionary apps find one word among millions in a few steps by halving the search.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 One of 8 coins is heavier. You use a balance and split the coins in half each time. How many weighings do you need?



a) 2 **b)** 3 **c)** 4 **d)** 8

2 You look for a name on 20 cards one by one. What is the most cards you might check?

3 How many 'Is it more than ...?' questions do you need, at most, to find a number from 1 to 16 by halving?

4 Look at the example strip. Which numbers are still possible?

5 A 2 kg bag is filled with 250 g packets. How many packets?

Now you build it

Write the 'guess my number' program using halving.

Set low to 1 and high to 32 **Repeat until found** **guess the middle of low and high**

If too small: set low to guess + 1 **If too big: set high to guess - 1**

Play with a friend. Count your guesses. Is it ever more than 6?

© 2026 Bullseye Knowledge System Limited

★ Challenge

One of 9 coins is heavier. What is the fewest weighings on a balance to be sure which one?

✓ How did I do?

I can search one by one.	I can search by halving.	I can explain which strategy is faster.
		

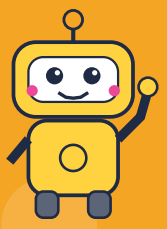
© 2026 Bullseye Knowledge System Limited

CHAPTER 9 · LEVEL B5

9 Coconut Farm

Coder skill: Input-output rules (function machines)

Goes with NCERT maths, Class 5, chapter 9

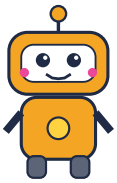


Learn

A **function machine** takes an **input**, follows a **rule**, and gives an **output**.

If you know some inputs and outputs, you can work out the hidden rule. Machines can be **chained**: the output of one becomes the input of the next.

function machine	rule	chain
input, rule, output	what the machine does	join machines one after another



Put a number into my machine, I follow my rule, and out comes the answer. Can you guess my secret rule?

Example

The harvest calculator

1. Each coconut tree gives 12 coconuts. Rule: **x 12**.
2. Input: 6 trees. Output: 6 x 12 = **72 coconuts**.
3. Change the input and the same rule gives a new output.



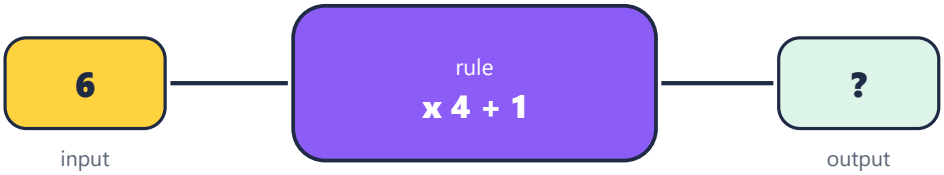
★ *Coder fact!* Every key on a calculator is a small rule machine: input in, rule applied, output out.

© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 What comes out of this machine?



2 Find the rule: 2 gives 10, 3 gives 15, 5 gives 25.

a) + 8 **b)** x 5 **c)** x 2 + 6 **d)** + 20

3 Using the rule **x 4 + 1**, which input gives an output of 49?

4 A farm has 25 trees with 30 coconuts each. How many coconuts in all?

5 The 750 coconuts are packed 12 to a bag. How many full bags, and how many coconuts are left over?

Now you build it

Design your own two-step machine and test it on a friend.

Ask for an input **step 1: ___** **step 2: ___** **say the output**

Give your friend three inputs and outputs. Can they find your rule?

© 2026 Bullseye Knowledge System Limited

★ Challenge

Chain two machines: **x 2** then **+ 3**. What does 7 give? Is it the same if you do **+ 3** first, then **x 2**?

✓ How did I do?

I can use an input-output rule.	I can find a hidden rule.	I can chain two machines.
		

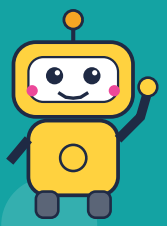
© 2026 Bullseye Knowledge System Limited

CHAPTER 14 · LEVEL B7

14 Maps and Locations

Coder skill: Capstone 1: design a treasure-map game

Goes with NCERT maths, Class 5, chapter 14

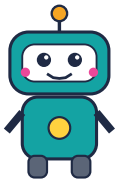


Learn

Maps use **grid references** (like C3) and **compass directions** (North, South, East, West). A **scale** tells real distance: 1 square = 100 m.

A good game has a **goal**, a **start**, **rules** and a way to **win**. Designing one uses everything you have learnt: routes, conditions, variables and testing.

grid reference	scale	game design
a letter and number for a square	map distance to real distance	planning goal, rules and winning

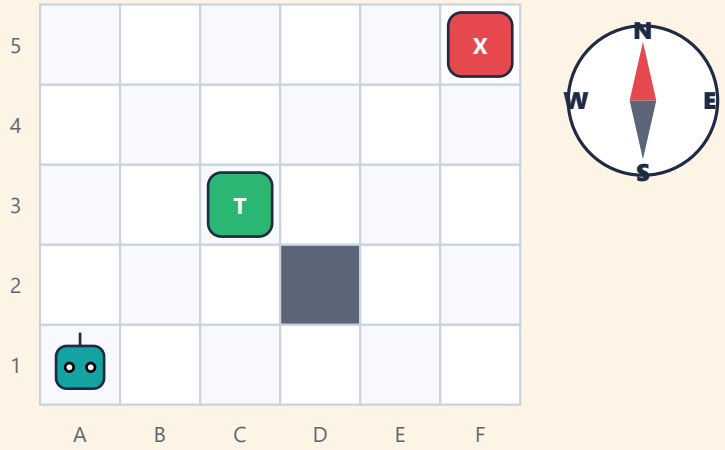


Let's make our own game! A map, a start, a treasure, some walls and clear rules. Then players write programs to win.

Example

The treasure map

1. Start at A1. The treasure X is at F5. The grey square is a wall. The tree T is at C3.
2. One route: North 4, then East 5. That is 9 moves.
3. Scale: 9 squares x 100 m = 900 m of walking.



© 2026 Bullseye Knowledge System Limited

Try it!

Circle the right answer, or write it in the box.

1 On the example map, what is at C3?

a) the treasure **b)** a tree **c)** a wall **d)** nothing

2 From A1, go 4 squares East and 4 squares North. Which square are you on?

3 1 square = 100 m. How far is it from A1 to F1 along the bottom row?

4 Which direction is the opposite of North-East?

a) South-East **b)** South-West **c)** North-West **d)** West

5 What is the fewest number of moves from A1 to the treasure at F5 on the example map?

Now you build it

Design your own treasure-map game on a 5 x 6 grid.

Choose a start square **Hide the treasure** **Add 2 walls and 1 bonus square**

Write the rules **Test it yourself**

Draw your map and write your rules. Test it: can the treasure always be reached?

Challenge

Write the shortest program for your own map. Ask a friend to find an even shorter one.

© 2026 Bullseye Knowledge System Limited

✓ How did I do?

I can use grid references and directions.	I can use a map scale.	I can design a game with clear rules.
		

© 2026 Bullseye Knowledge System Limited

Sample teacher script: Guess my number: halving search

From the teacher book, Class 5, chapter 8 (*Weight and Capacity*). One 40-minute period. Level B6.

You need	
Kit	Number strip 1-32 (floor or board), 8 coin counters, a simple balance or its picture, search code cards
Book	Chapter 8 pages, 'Try it' questions 1-4
Goal	Children compare one-by-one search with halving search, count the guesses and explain why halving is faster

Step and time	Teacher does	Children do
1. Predict and run (8 min)	Think of a number from 1 to 32. Let the class guess 1, 2, 3, ... in order.	Predict the most guesses this could take (32). Then watch the teacher guess the middle each time instead.
2. Investigate (8 min)	Ask: 'How many numbers did one guess rule out? Why is the middle the best guess?'	Discuss in pairs: each middle guess rules out half of what is left.
3. Modify (8 min)	Change the range to 1-64.	Predict how many more guesses are needed (just one more split).
4. Put in order (8 min)	Hand out the jumbled halving-search code cards.	Order them into a working program, then play one round in pairs.
5. Make (8 min)	Show 8 coins, one secretly heavier.	Plan weighings that halve the coins each time; find the heavy coin in 3 weighings.

Lesson set-up



Set-up: a number strip on the board (shown here for 1-16: guess 8, then 12). Grey out each half as it is ruled out, so children see the range shrink.

Common mistakes to watch for

- **Guessing the middle of the whole strip again** instead of the middle of what is left.
- **Off by one:** forgetting that the guessed number itself is also ruled out.
- **Thinking halving always wins.** For 3 cards, checking one by one is just as quick; halving shines for long lists.

© 2026 Bullseye Knowledge System Limited

Different paces

Support: use the range 1-16 and grey out each half on the strip.
Stretch: how many guesses for 1-100? For 1-1,000? (about 7 and 10)

Check for understanding

Two quick questions at the end of the period:

1. For 1-64, at most how many 'Is it more than ...?' questions are needed? (6) 2. One of 8 coins is heavier. How many weighings with a balance? (3)

Note in the observation journal: **Beginning** (searches one by one), **Developing** (halves with help), **Secure** (halves the remaining range alone), **Extending** (predicts the number of guesses for a new range).

Curriculum link

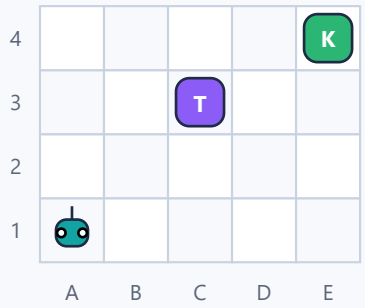
CBSE's Class 5 handbook compares searching one by one with divide-and-conquer using a weighing scale in *Weight and Capacity*. This lesson makes halving explicit: it is binary search, one of the most used ideas in computing.

Teacher notes

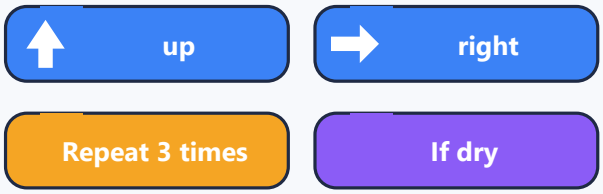
© 2026 Bullseye Knowledge System Limited

The classroom kit

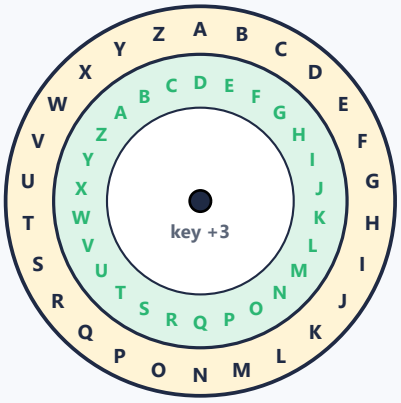
One kit per class section, used all year. Everything is printed, laminated and reusable.



Floor grid mat (6 x 6 ft) for robot routes



Code cards: arrows, repeat, if, when (4 decks)



Cipher wheels for secret codes (20)



Picture and shape cards for patterns and sorting (120)

Item	Quantity per kit
Code card decks (playing-card size, laminated)	4 decks of 54
Cut-out coding blocks (die-cut sheets)	20 sheets
Floor grid mat, washable	1
Cipher wheels	20
Picture and shape cards	120
Robot direction tokens	30
Teacher quick-guide card	1
Sturdy box	1

Illustrations show the design. Final materials will be confirmed after the pilot.

© 2026 Bullseye Knowledge System Limited

Assessment and the report card

CBSE asks for continuous, competency-based assessment in CT&AI. The teacher book includes an observation journal using four levels, and a term summary that fits on the report card.

Skill	Beginning	Developing	Secure	Extending
Following steps	Follows short steps with help	Follows and fixes simple steps	Writes steps with loops	Designs and improves own steps
Patterns	Spots simple repeats	Continues a pattern	Finds the core and rule	Makes new patterns and explains them
Breaking problems down	Needs help to start	Breaks familiar jobs into steps	Breaks new problems into steps	Plans steps and their order alone
Logic and rules	Uses one clue	Uses two clues or an if-then rule	Uses several clues and if-else	Spots extra clues and explains

Sample term summary

Area	Level	Teacher comment
Computational thinking (overall)	Secure	Breaks problems into steps and writes loops confidently.
Coding on paper	12 of 14 'build it' tasks	Ready for if-else rules with more than two choices.
Self-check ('How did I do?')	Mostly green	Finds codes easy and clues harder; we practised clue puzzles together.

Children check themselves too

Each chapter ends with **'How did I do?'**, where children colour a face for each 'I can' statement. Teachers can compare this with what they observed, which helps children learn to judge their own progress.

© 2026 Bullseye Knowledge System Limited

Answers for the sample chapters

8. Weight and Capacity

1. b) 3 (4 vs 4, then 2 vs 2, then 1 vs 1)
2. 20
3. 4
4. 17 to 23
5. 8
Build it: No: halving finds any number from 1 to 32 in at most 6 guesses (5 yes/no splits)
Challenge: 2: weigh 3 against 3; the heavy group (or the 3 left out) is known; then weigh 1 against 1 from that group

9. Coconut Farm

1. 25
2. b) x 5
3. 12
4. 750
5. 62 bags, 6 left over
Build it: Open answer
Challenge: 17; no: + 3 first gives 20, so the order matters

14. Maps and Locations

1. b) a tree
2. E5
3. 500 m
4. b) South-West
5. 9
Build it: Open answer
Challenge: Open answer

© 2026 Bullseye Knowledge System Limited

Cut-out code cards

Ask an adult to help you cut along the dotted lines. Use the cards to build programs on your desk, then act them out with a friend.

up	down	left
up	down	left
right	Start	Repeat ___ times
right	Start	Repeat ___ times
If ___	else	When ___ happens
If ___	else	When ___ happens
add ___	take away ___	Stop
add ___	take away ___	Stop

© 2026 Bullseye Knowledge System Limited

A note for parents

Schools can send this page home at the start of the year.

Dear parents,

This year your child will learn **computational thinking** as part of maths, following CBSE's new curriculum for Classes 3-8. Computational thinking means solving problems the way a programmer does: breaking a big job into steps, spotting patterns, keeping what matters and following clear rules. These skills help in maths, reading and everyday life, and they are the base for coding and AI later.

Your child will use the *Think Like a Coder* workbook in class. There are no screens: children solve puzzles, crack secret codes, guide a robot around a grid and write programs with paper code cards.

How you can help at home

- ✓ Ask your child to explain a puzzle to you. Explaining is part of the learning.
- ✓ Play 'robot' at home: your child gives you step-by-step instructions to make a sandwich, and you follow them exactly.
- ✓ Spot patterns together: on clothes, tiles, calendars and number plates.
- ✓ Praise effort and good thinking, not just right answers.

Take-home challenge

Write the numbers 1 to 20. Colour every 3rd number. What pattern do you see? Which number would be the 10th coloured one?

Answer: 3, 6, 9, ... the 10th is 30.

No child's personal data is collected by the workbook. Thank you for your support.

Class teacher _____ **School** _____

© 2026 Bullseye Knowledge System Limited

Next steps for your school

1. **Look through this sample**
   Try a chapter with a class or ask a teacher to review it.

2. **Talk to us**
   A short meeting with the principal and coordinator to answer questions.

3. **Start a free pilot**
   Two or three sections, January to March 2027, with an orientation and weekly support.

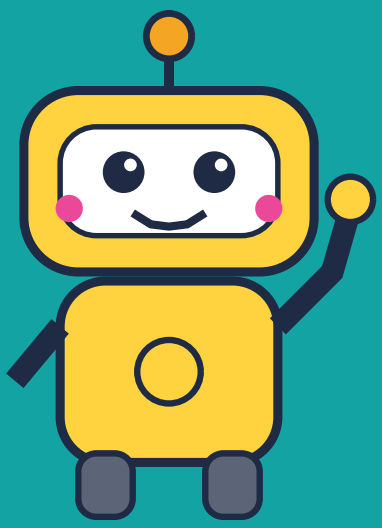
4. **Add it to the 2027-28 book list**
   Most schools finalise their list by the end of January.

Contact

Anoop Dobhal, CodeSchool, Bullseye Knowledge System Limited
Phone / WhatsApp: [Phone] · Email: [Email]

Notes

© 2026 Bullseye Knowledge System Limited



Think Like a Coder

Computational thinking for Classes 3-5, for CBSE schools

Ask us about a free pilot

Phone / WhatsApp: [Phone] · Email: [Email]

© 2026 Bullseye Knowledge System Limited. All rights reserved. Specimen copy, not for sale.
CodeSchool is a working name. Not an official CBSE or NCERT publication.